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**Comparative Analysis of Classification Techniques in Data Mining:
K-Nearest Neighbors, Naive Bayes, and Decision Trees based Approach**

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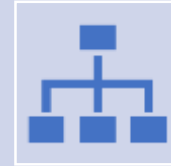
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Introduction:



Data mining is crucial for extracting insights from large datasets.



Classification is a key task with applications in healthcare, finance, etc.



Three classical algorithms widely used: KNN, Naive Bayes, Decision Trees.



Goal: Compare their performance on a heart disease dataset.

Relative Works

- **KNN:** Instance-based, good for nonlinear boundaries, but slow with large data.
- **Naive Bayes:** Probabilistic, fast, works well with high-dimensional text data.
- **Decision Trees:** Rule-based, interpretable, but prone to overfitting.

Research Gap: Most studies focus only on accuracy, not on interpretability, speed, or medical applicability.

Method	Strengths	Weaknesses
KNN	Flexible, no training needed	Slow prediction, sensitive to noise
Naive Bayes	Fast, handles high dimensions	Assumes feature independence
Decision Tree	Interpretable, handles mixed data	Overfitting, unstable

Research Questions



Which classifier achieves the highest **accuracy** for heart disease prediction?



Which model best balances **precision and recall** (F1-Score)?

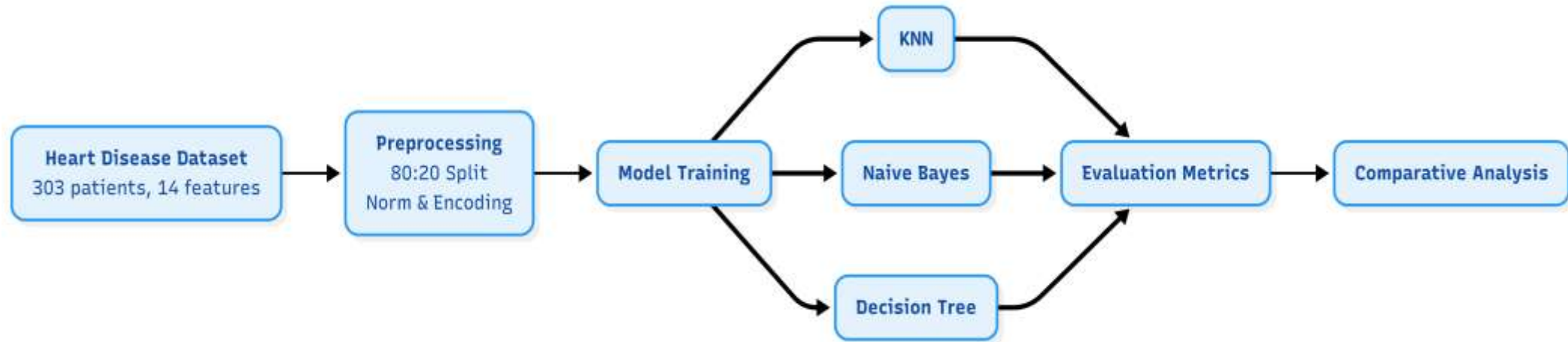


How does **computational efficiency** compare across models?



Which algorithm is most **interpretable** for clinical use?

Methodology:



Dataset: 303 patient records with 14 clinical features.

Processing: Features normalized and split into 80% training / 20% testing sets.

Models: Implementation of KNN, Naive Bayes, and Decision Tree algorithms.

Metrics: Comparative analysis based on classification accuracy and performance.

This systematic framework ensures robust heart disease prediction by benchmarking multi-model performance against preprocessed clinical data.

Dataset & Preprocessing

Dataset: UCI Heart Disease (303 instances, 14 attributes)

Features: Clinical (chol, bp), demographic (age, sex), test results (thalach, oldpeak)

Preprocessing:

- Min-Max Normalization (KNN)
- Standardization (Naive Bayes)
- Train-Test Split: 80%-20% stratified

Models Implemented

1

KNN: Tuned k (5-29), distance-based, normalized.

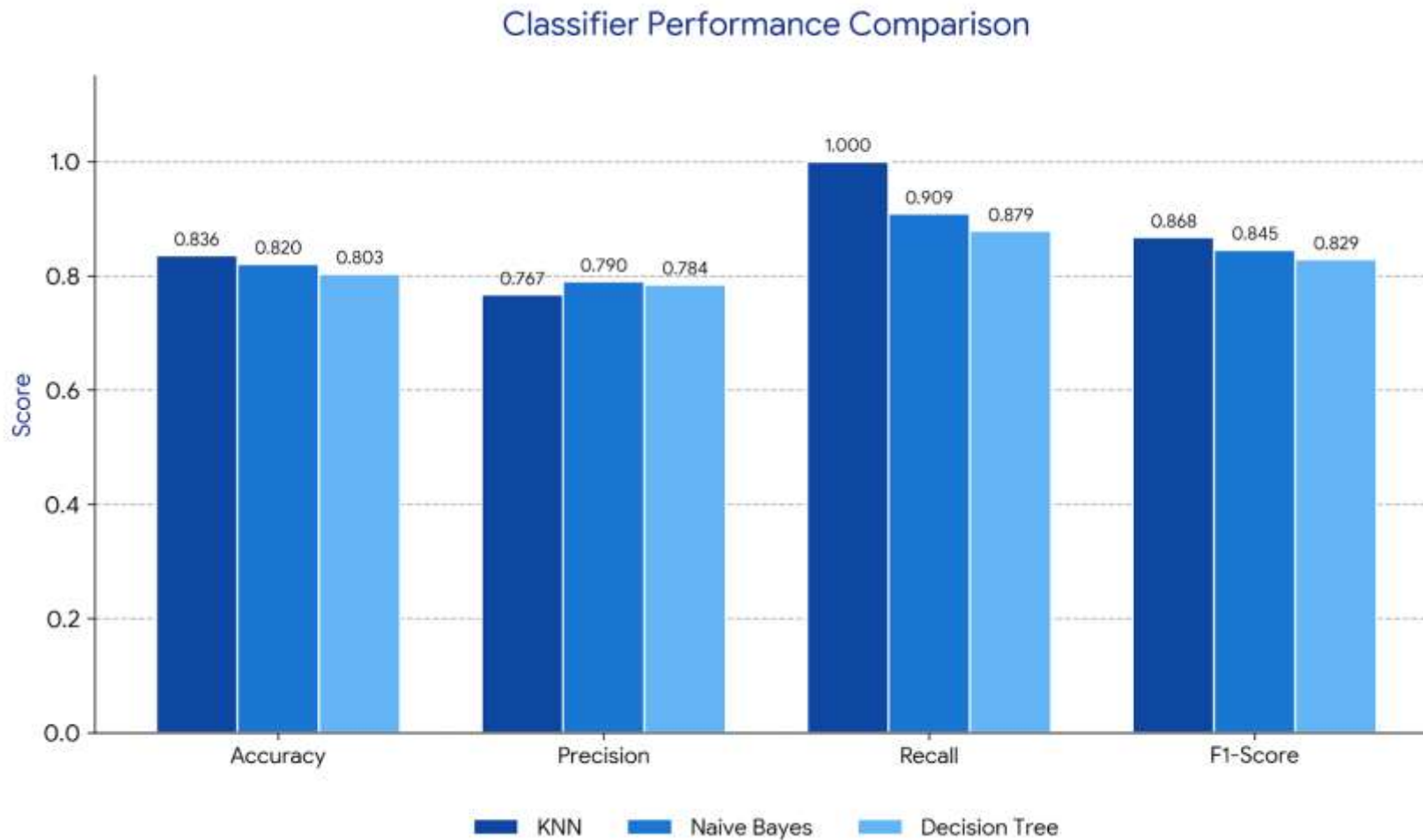
2

Gaussian Naive Bayes: Probabilistic, assumes normal distribution.

3

Decision Tree: Gini/Entropy split, depth tuning (2–10), no scaling needed.

Results - Performance Metrics

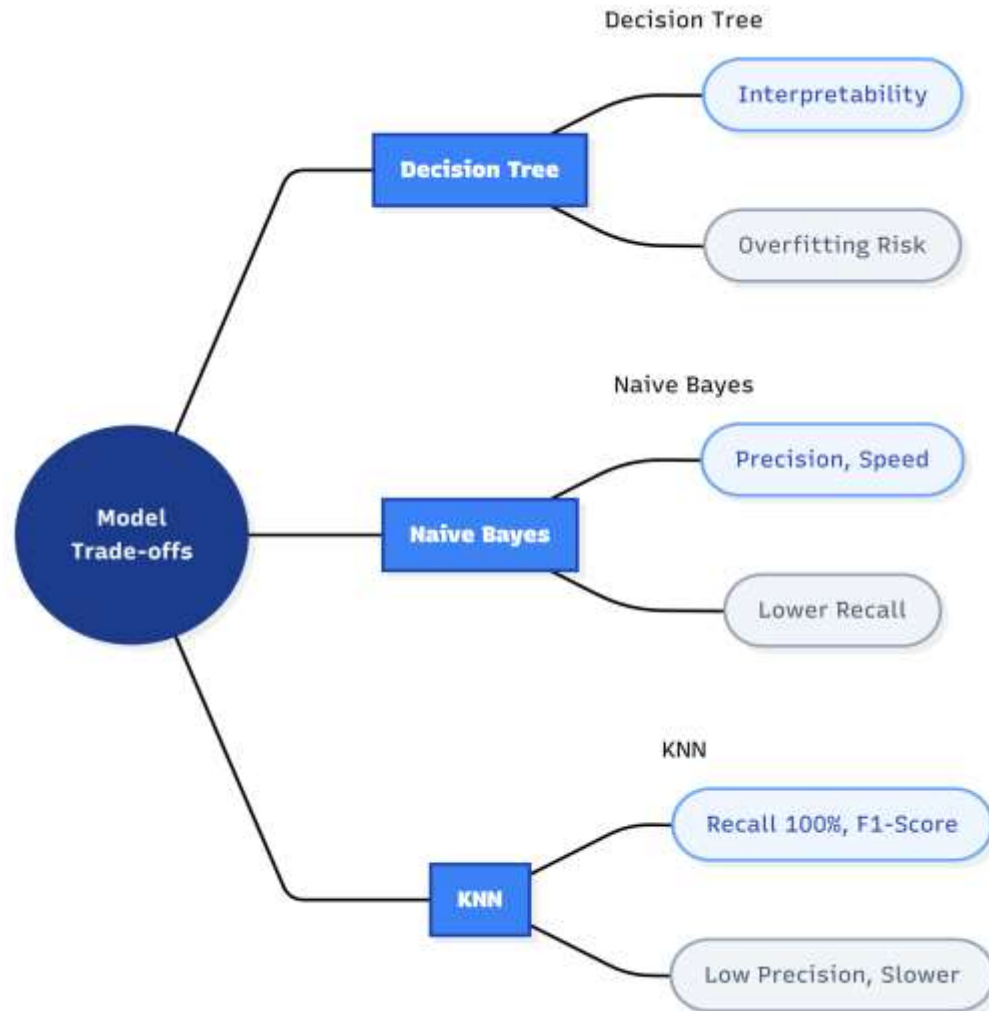


KNN: Best Accuracy & Recall

NB: Best Precision

DT: Lowest overall but interpretable

Results - Comparative Analysis



- **Trade-offs Summary:**

Model	Best At	Weakness
KNN	Recall (100%), F1-Score	Low Precision, Slower Prediction
Naive Bayes	Precision, Speed	Lower Recall
Decision Tree	Interpretability, No Scaling	Overfitting, Lower Accuracy

Significance of the study



Provides **evidence-based guidance** for choosing classifiers in medical diagnostics.



Highlights **trade-offs between sensitivity and precision** in clinical settings.



Emphasizes **model interpretability** for clinician trust.



Offers a **reproducible framework** for benchmarking classifiers on health data.

Conclusion & Model Selection Strategy



KNN: The Gold Standard for Safety - With **100% Recall**, it is the optimal choice for clinical screening where missing a diagnosis is the highest risk.



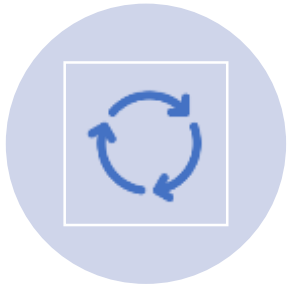
Naive Bayes: Optimized for Efficiency - Best suited for real-time applications requiring high **Precision** and minimal computational latency.



Decision Tree: Transparent Diagnostics - Most valuable in clinical settings where Interpretability and understanding "why" a diagnosis was made are essential.

There is no 'universal' best classifier; model selection must be dictated by clinical priorities and balancing the need for diagnostic sensitivity against operational speed and transparency.

Future Work



Extend to **ensemble methods** (Random Forest, XGBoost, SVM).



Test on **larger, imbalanced datasets**.



Incorporate **feature selection & hyperparameter optimization**.



Explore **explainable AI (XAI)** for clinical adoption.

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Thank You for your attention!

Any Questions?

